

Part D – Reflection Questions

1. Which wireless/WAN technology choice in your design was hardest to justify, and why?

The hardest technology to justify was the use of VSAT satellite technology for global link. VSAT can provide connectivity to remote locations where fibre or other terrestrial connections may not be available, making it suitable for areas with limited infrastructure. However, it can be expensive to install and maintain and may have higher latency than fibre. We therefore had to consider the location, reliability, coverage and cost before choosing it for the global connection.

2. How does Namibia's geography and population density shape the WAN/wireless choices you made for the Windhoek–branch link, compared to a similar link in a densely populated country?

Namibia has a large geographical area with a relatively low population density, meaning that some branches can be located far away from major cities and existing network infrastructure. Because of this, technologies such as microwave wireless links, VSAT and other wireless solutions can be useful where laying fibre would be expensive or impractical. In a densely populated country, fibre would usually be easier to deploy because there are more customers and existing infrastructure to support it. Our design therefore considered Namibia's distances, terrain and limited infrastructure when selecting WAN technologies.

3. What cross-border challenges did you identify for the SADC link, and how might regional cooperation reduce these?

Some cross-border challenges include different telecommunications regulations, licensing requirements, network standards, infrastructure availability and costs between countries. There can also be challenges with coordinating network providers and maintaining reliable connectivity across borders. Regional cooperation within SADC could help by creating common regulations and standards, encouraging shared infrastructure and making cross-border telecommunications licensing easier. This would improve connectivity and potentially reduce costs for businesses operating across multiple countries.

4. How did the global case study you researched change your understanding of what is technically possible versus what is currently deployed in Namibia?

The global case study showed us that many advanced technologies are technically possible, including high-speed fibre networks, advanced wireless systems, satellite connectivity and cloud-based networking. However, not all of these technologies are widely deployed in Namibia because of factors such as cost, population density, infrastructure availability and geographical challenges. This helped us understand that designing a network is not only about choosing the fastest technology, but also about selecting a solution that is practical and affordable for the environment where it will be deployed.

5. What did using GitHub as a team teach you about version control and collaborative workflows in a real IT project?

We learned how to upload files, make changes, and keep track of what was changed and who made the changes (version control). This was useful because if something went wrong, we could see the previous version and understand what had been changed. We also learned that it is important to commits (give record of changes) clear descriptions so that other team members can understand what was done.

We also learned that communication is key when working together as a team. Before manipulating, changing or replacing files, team members need to know what others are working on. This helps prevent conflicts where two people make changes to the same file, and one person's work from getting overwritten.

GitHub gave us an overall better understanding of how real IT teams work together on projects. It taught us that collaboration is not just about sharing files, but also about keeping the projects organised, tracking changes, communicating with team members, and making sure everyone can work on the same project without losing each other's work.

6. If Kalahari Logistics doubled its number of branch offices next year, what would you change about your design to scale it?

In the event of Kalahari Logistics doubling their number of branch offices next year, we would need to design the network to make it easier and more scalable for future additions so that adding new branches does not require us to redesign the whole network, every time.

Firstly, we would use another IP addressing plan. In order to prevent IP addressing conflicts we will assign dedicated addresses instead of choosing addresses randomly, to the different branches, this also makes it easier to assign IP addresses to new offices. we would reserve a separate IP range for each branch.

Secondly, we would also move from static routing to dynamic routing, such as OSPF. With our current design, every new branch would require routes to be manually added to

the routers. With dynamic routing, the routers can automatically learn about new networks, which would save time and reduce configuration mistakes.

Thirdly, For the connecting between branches and the main office, we could use site-to-site VPNs over the Internet. This would be more practical and less expensive than having a separate dedicated connection for every branch. For branches in areas where fibre or reliable Internet is not available, microwave or satellite connections could be used instead.

We would also look at the bandwidth requirements at the Windhoek headquarters. The main connection might need to be upgraded, because more branches would mean more users, more Internet traffic and more data being sent to the head office. Compensation is required.

Last, not least, we would consider centralized network monitoring and management. This would allow the IT team to monitor all the branches from one centralized location to quickly and efficiently identify problems for mitigation, recovery and management.

The main change would be to design the network so that new branches can be added easily without having to rebuild or reconfigure the entire network. This would make the network easier to manage, more reliable and ready for future growth.

7. Most Valuable Lesson and Connection to a Networking Career

The most valuable lesson our group learned from this assignment is that networking is not just connecting routers, switches and computers together. There is a lot more to consider when designing a network for a real company. We had to consider things like cost, distance, reliability, available technology, security and scalability. For instance, a connection that works well in Windhoek headquarters might not be the best option for a branch in a remote area where fiber is not available. In that situation, technologies such as wireless, microwave or satellite may be more suitable.

We also learned that planning is very important. Before building the network, we needed to decide how the different sites would connect, how IP addresses would be allocated and how the routers would communicate. Using Cisco Packet Tracer gave us practical experience with configuring devices, testing connections with command prompt and traceroute and troubleshooting problems when something did not work. This was especially useful because we had to find and fix configuration issues instead of simply following theory.

Another important lesson was working as a team. Using GitHub showed us how important it is to keep everyone's work organized when several people are working on the same project. We learned about version control and commits, which allowed us to keep track of changes and see what each team member had contributed. It also teaches us that

communication is important because team members need to know what others are working on to avoid overwriting or losing work.

The assignment gave us an overall better understanding of what a real networking job involves. A network professional needs to be able to plan a network, choose the right technology, configure and troubleshoot equipment, document the work and work together with a team. These are skills that we will need in a real IT or networking career, where the goal is not just to make a network work, but to make sure it meets the business requirements and can continue operations as the company expands.

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